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The Effect of injectable platelet-rich fibrin and low-level laser therapy on graft shrinkage in free gingival grafts: a randomized controlled clinical trial

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Abstract

Background The aim of this randomized, controlled clinical study was to evaluate and compare the effects of injectable platelet-rich fibrin (i-PRF) applied over free gingival grafts (FGG) on graft shrinkage, root coverage, and postoperative comfort, in comparison with low-level laser therapy (LLLT) and a control group.

Methods A total of 48 patients aged 19–41 years with Cairo et al. Gingival Recession Type 1 (RT1) were randomly allocated into control ($n=16$), i-PRF ($n=15$), and LLLT ($n=17$) groups. The primary outcome was graft shrinkage. Secondary outcomes included gingival recession depth (RD), keratinized tissue height (KTH), probing depth (PD), clinical attachment level (CAL), and postoperative outcomes (oedema, erythema, pain, and analgesic consumption). LLLT was applied immediately after surgery and on days 3, 7, and 14 using a 940 nm diode laser (0.21 W, 5 J/cm^2). The i-PRF was prepared by centrifuging venous blood at 700 rpm for 3 min and was then applied in gel form onto the graft surface. Graft areas were measured at baseline and at 1, 3, and 6 months using image analysis software. Postoperative oedema, erythema, pain, and analgesic consumption were also evaluated.

Results Graft shrinkage was significantly higher in control subjects than in both treatment groups, and significantly higher in LLLT than i-PRF at all timepoints. Postoperative edema, erythema, and VAS scores showed no significant intergroup differences. Analgesic use was higher in the control group than in i-PRF. KTH values were greater and RD values lower in the i-PRF group compared with the control and LLLT groups.

Conclusion i-PRF reduced graft shrinkage, enhanced root coverage, and improved postoperative comfort, whereas LLLT contributed minimally to graft stability. i-PRF appears to be an effective biological adjunct in FGG surgery.

Trial registration Retrospectively registered at ClinicalTrials.gov (Identifier NCT07303192 Date of registration 20251211).

Keywords Free gingival graft, Injectable platelet-rich fibrin, Shrinkage, Low-level laser therapy

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Introduction

Gingival recession (GR) is defined as exposure of the root surface through a change of the gingival margin towards the apical of the cement-enamel junction (CEJ) line. This can lead to aesthetic problems, dentin sensitivity and loss of periodontal tissues [1]. A sufficient width of adherent gingiva is important for the preservation of periodontal health. Some researchers have reported that at least 2 mm keratinized gingiva is required [2, 3], of which 1 mm is adherent, while other studies have shown that a narrower keratinized tissue width could be sufficient to obtain effective plaque control [4, 5]. However, thin gingiva phenotype or the presence of insufficient keratinized tissue makes plaque control more difficult, and increases the risk of inflammation and attachment loss [6].

Free gingival graft (FGG) is a method widely used to increase keratinized tissue width [7]. However, dimensional shrinkage of the graft during the healing process is a significant disadvantage, and rates reported in the literature vary between 12% and 58% [8]. Minimizing graft shrinkage is clinically important, as excessive shrinkage may compromise the final keratinized tissue width and adversely affect long-term soft tissue stability and periodontal health [7–10]. Factors affecting the development of vascular connections between the graft and the recipient region play a determinant role in graft shrinkage [11]. Therefore, the support of collateral circulation is important. In previous studies, different methods have been used to reduce graft shrinkage, such as the application of cyano-acrylate [12], Low-Level Laser Therapy (LLLT) [7, 13], or hyaluronic acid [14].

LLLT, which is usually applied with diode lasers, is a non-invasive method working with biomodulator effects rather than creating heat. Although the mechanism is not fully understood, it has been shown to accelerate cell division by changing the cellular metabolism, increasing fibroblast migration, and supporting matrix production [15]. It has also been suggested that it accelerates wound healing by promoting angiogenesis, collagen synthesis, and the formation of epithelial tissue with fibroblasts, which supports regeneration by regulating the metabolism of surrounding tissues [16, 17]. In a previous study, less postoperative pain and less shrinkage in FGG was obtained with the use of 810 nm diode laser [7]. However, there are few studies in the literature that have evaluated the efficacy of LLLT on FGGs.

Platelet-rich fibrin (PRF), which contains high concentrations of growth factors such as platelet-derived growth factor (PDGF), vascular endothelial growth factor (VEGF), and transforming growth factor- beta (TGF-β), is a plasma fraction that provides long-term expression of these factors due to the fibrin matrix [18–20]. One of the most important advantages of PRF is that immune-compatible growth factors can be obtained at low cost without the use of anticoagulants [19]. Standard PRF formulations do not contain fluid, and the growth factors are encapsulated within the fibrin matrix. Therefore, injectable PRF (i-PRF) was developed as a liquid formulation of PRF [21]. i-PRF differs from other PRF types as it can be obtained in liquid form which rapidly transforms into a gel. Injectable platelet concentrates such as i-PRF provide higher cell migration and greater total long-term growth factor release compared to membrane-based platelet concentrates [22]. i-PRF increases cell migration, and significantly increases the expression of type 1 collagen with TGF-β, PDGF, and osteocalcin mRNA. Within a gel-like three-dimensional fibrin network, i-PRF contains thrombocytes, leukocytes and growth factors, and provides expression of these factors for up to 10 days [21]. Due to these properties, it is thought to support wound healing and contribute to graft stability. Although a previous study has reported that in coronally advanced flap (CAF) and combined connective tissue graft (CTG) the application of i-PRF reduced the depth of recession and increased keratinized tissue height [18], no study could be found in the literature that has evaluated the application of gel i-PRF on FGG. In another study, the application of i-PRF to the root surface for biomodification was reported to result in favorable clinical outcomes [23]. However, this approach does not directly target the graft tissue. Therefore, the hypothesis of the present study was that the application of i-PRF directly onto the graft surface, following its transformation into a gel form after preparation, may enhance the local retention of growth factors and thereby more effectively support graft healing.

The aim of this study was to evaluate the effects of gel-form i-PRF application on FGG in respect of graft shrinkage, root surface coverage rate, and postoperative patient comfort, and to compare these effects with an LLLT-treated group and a control group. The null hypothesis was that neither gel i-PRF application nor LLLT application would lead to a significant difference in graft shrinkage, root surface coverage rate, and postoperative patient comfort compared to the control conditions.

Material and Method

Study population

This controlled, randomized, parallel-group study with a researcher-blinded design was conducted between 1 July 2024 and 30 October 2025. Approval for the study was granted by the Clinical Research Ethics Committee of Inonu University (protocol no: 2024/84) and all procedures were in compliance with the 2013 updated principles of the 1975 Helsinki Declaration. The study population consisted of 48 patients with Cairo et al. Gingival Recession Type 1 (RT1) who were treated in the Periodontology Department of Kahramanmaraş Sütçü

Imam University Dentistry Faculty. Written and verbal consent was obtained from all the study participants. The study protocol was registered on ClinicalTrials.gov (Identifier: NCT07303192; Date of registration: 2025-12-11), and the reporting process was carried out in accordance with the principles outlined in the CONSORT 2025 statement [24] (Fig. 1).

Study inclusion criteria

The patients included in the study were aged > 18 years, were systemically healthy, were determined with RT1 [25] in vital incisor teeth in the mandibular anterior region, and no caries or restorations in the relevant teeth. Following initial periodontal therapy (INT), patients presenting with full-mouth plaque scores (FMPS) and full-mouth bleeding scores (FMBS) below 20% were included in the surgical phase [26]. In addition, based on the post-INT evaluation, only teeth with plaque index (PI) and gingival index (GI) scores below 1 were included in the study [7]. The included teeth exhibited a keratinized gingival height of less than 2 mm.

Study exclusion criteria

Patients were excluded from the study if they had active periodontal disease or probing depth >3 mm [7], if they were using any drugs that could prevent surgery or affect wound healing, if they were pregnant or breastfeeding, had a history of periodontal surgery in the working area, were active smokers, or were currently undergoing orthodontic treatment.

Sample size calculation

Sample size analysis was performed using G*Power 3.1.9.4 (Franz Faul, Universität Kiel, Kiel, Germany) in the framework of repeated-measures ANOVA (within–between interaction), based on graft shrinkage as the primary outcome [7]. Assuming a medium-to-large effect size ($f = 0.30$), $\alpha = 0.05$, power $(1 - \beta) = 0.80$, four repeated measurements, a correlation among repeated measures of 0.50, and a non-sphericity correction (ε) of 0.70, the required minimum total sample size was calculated to be 33 participants (11 per group). As this study included three groups with a total of 48 participants, the achieved sample size exceeds the minimum requirement, indicating that the study is sufficiently powered to detect group-by-time interaction effects.

Randomization and blinding

Using a computer-generated randomization table the patients were assigned to one of three treatment groups. The randomization sequence was generated by an independent individual who was not part of the research team and had no involvement in the study. Each patient participated in the study with a single recession defect which affected the inferior incisor tooth. Blinded group allocation was achieved using sealed, opaque envelopes containing the treatment to be applied. These sealed

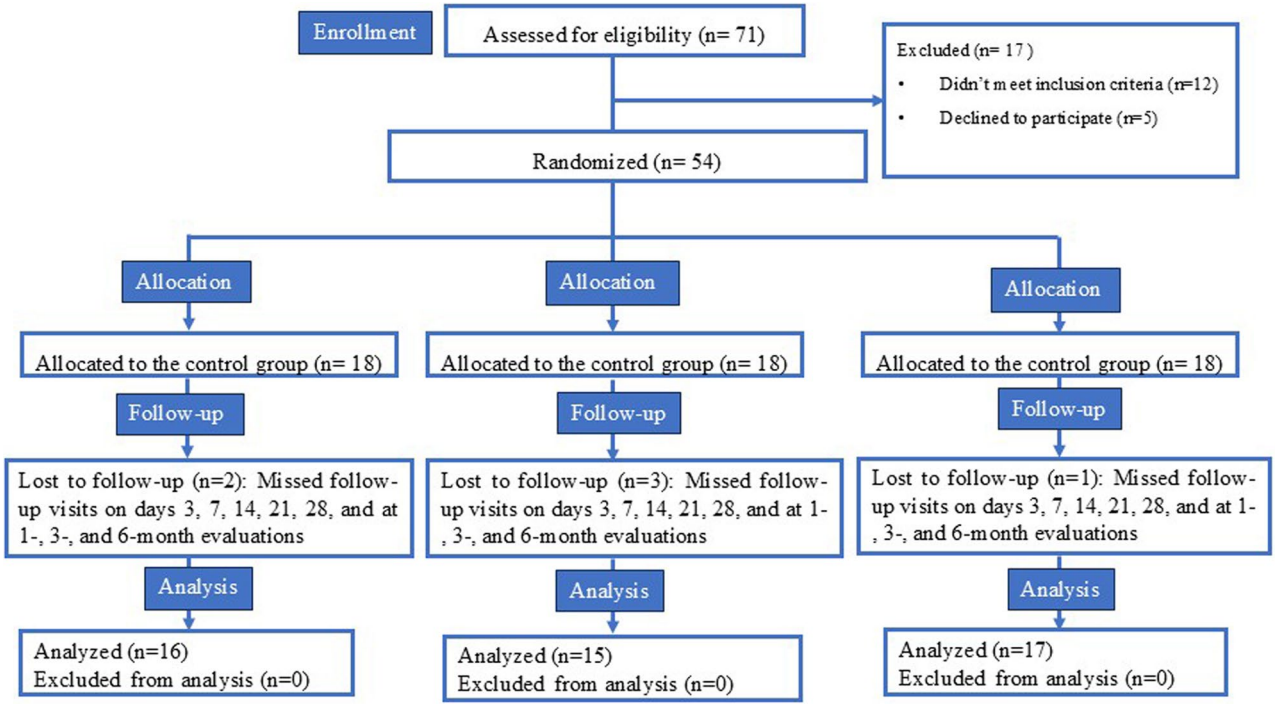


Fig. 1 Study flowchart

envelopes containing the treatment allocation were opened immediately after the graft harvesting procedure was completed.

All surgical procedures and LLLT application in the relevant group were performed by an experienced periodontist (E.Ç.Ö.). All clinical measurements and evaluations were performed by a single researcher (E.B.) who was blinded to the patient group allocation and treatment type. Photographs and data were coded to prevent identification of the treatment groups.

Presurgical treatment

All the patients underwent primary periodontal treatment and professional mechanical plaque cleaning was performed according to the new periodontal and peri-implant diseases and conditions classification guidelines of the American Periodontology Academy 2017 Working Group [27]. With evaluation of the baseline FMPS and FMBS, the percentage of positive surfaces was calculated using four sides per tooth. Then oral hygiene education and motivation was given to each patient. At three weeks after INT, patients who met the inclusion criteria were enrolled in the surgical phase.

Surgical procedure

Infiltration type anesthesia was applied to the region planned to be the surgical site (Ultracaine DS, Sanofi Aventis, Türkiye). When preparing the recipient area, starting from the enamel-cement junction of the tooth with recession, with a length of 2 mm on each side, two horizontal cuts were made in the mesial and distal directions without reaching the gingival edge of the adjacent teeth. Two vertical cuts were then made finishing 3–4 mm apical of the mucogingival line. The vertical cuts joined the horizontal cuts in the mucogingival line. Sharp dissection of partial thickness was performed apical of the alveolar mucosa and the epithelial surfaces within these cuts were removed [28]. Connective tissue over the existing bone was protected to increase blood support in the region. Root planning was performed on the root surfaces of the teeth to decrease the graft and avascular contact surface. The recipient area was washed with physiological saline and a moistened gauze pad was placed over it until the graft was placed.

After the application of local anesthesia, FGG was taken at least 2 mm from the gingival edge from the palatal region remaining between the distal surface of the canine tooth and the mesial surface of the first molar tooth. A partial thickness graft of 1.5–2 mm thickness was removed, formed from the epithelium and underlying fine connective tissue layer. After harvesting the FGG, the palatal wound was sutured (Pegelak®, 5–0 poly (glycolide-co-lactid) (PGLA) sutures, Dogsan A.S., Trabzon, Turkey) using a collagen sponge (Spongostan; Ethicon, Somerville, USA).

The FGG was adapted to the recipient region by correcting the graft contours, and was then fixed to the periosteum using simple interrupted sutures (Prolene®, Prolene 6 – 0, Ethicon, Johnson & Johnson, Rockville, MD, USA). By continuing along the lateral edges of the graft the suturing provided full stability. When necessary, vertically hanging cross stitches were made to provide slight tightening of the graft to the recipient area (2 cases in each group). No sutures were applied to the apical section of the mucosa. After graft placement, gentle pressure was applied for 2 min using saline-soaked gauze to ensure proper adaptation of the graft to the recipient bed and to prevent dead space and hematoma formation [7]. This procedure was performed prior to i-PRF/LLLT application and periodontal dressing.

Study groups

A total of 48 patients were included in the study in the three groups of control group (n = 16), i-PRF group (n = 17), and LLLT group (n = 15). A GaAlAs semi-conducting diode laser with wavelength of 940 nm, output power of 0.21 W, in continuous mode, spot size of 2.8 cm^2, and energy density of 5 J/cm^2 (Wiser Doctor Smile, Lambda SPA, Italy) was used in the LLLT group [29]. The laser was applied in a non-contact mode using a pain-relief handpiece, at a distance of 2–3 mm from the wound surface [29], with a 60-second application time [7], and the output power was calibrated by the manufacturer. This application was performed on the recipient area immediately after the surgical procedure and on postoperative days 3, 5, 7, and 14 [30]. In each session, the probe was positioned perpendicular to the wound surface to ensure uniform energy delivery, and the laser was applied to cover the entire recipient site [7].

In the i-PRF group, blood was withdrawn from the antecubital vein of the patients into 10 ml glass covered plastic tubes (Vacuette Tube, Greiner Bio-One) without anticoagulant. The i-PRF was prepared by centrifuging the blood samples at 700 rpm for 3 min (60 g) using a Medifuge device (Silfradent Medifuge MF200, Italy) [20]. After the centrifugation, the i-PRF formed in the upper part of the tube was collected in a syringe and transferred to a metal godet, then left for 15–20 min for the completion of polymerization. The i-PRF gel, obtained after polymerization, was applied directly to the FGG surface immediately following graft placement and suturing. It was gently spread to cover the entire graft area, and stabilization was achieved through contact with the recipient site natural tissues and the overlying periodontal dressing, without the need for additional fixation. (Fig. 2).

In the control group, no additional procedure was made on the FGG. In all three groups, the recipient

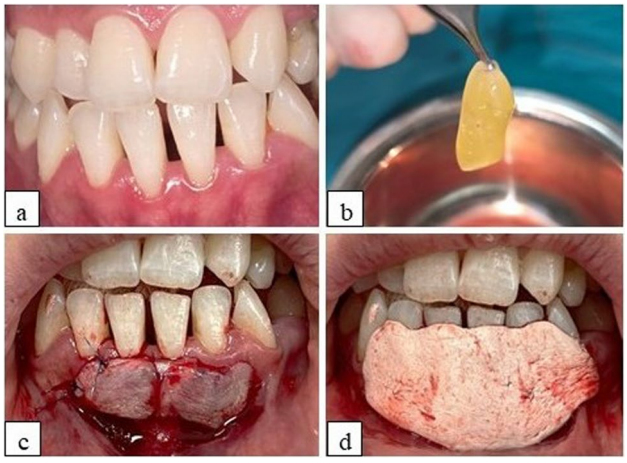


Fig. 2 The preparation of i-PRF and its application to the recipient site. **a** Appearance of the recipient site before FGG harvesting, (**b**) appearance of the polymerized i-PRF, (**c**) Application of the polymerized i-PRF onto the FGG, (**d**) Coverage of the recipient site with periodontal dressing

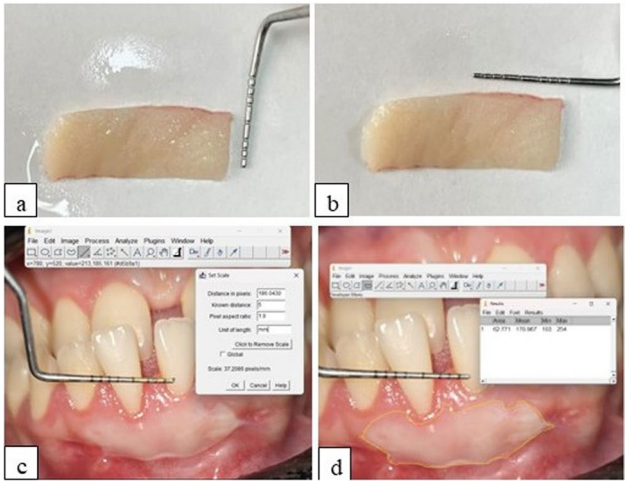


Fig. 3 Assessment of graft dimensions and FGG area. **a** Measurement of graft length, (**b**) Measurement of graft width, (**c**) Calibration of ImageJ NIH using a periodontal probe, (**d**) Delineation of the FGG area in the calibrated image

sites were covered with aluminum foil and a periodontal dressing (Coe-Pak, GC America Inc., IL, USA) for 3 days [31].

Postoperative care

The patients were instructed that they should avoid tooth brushing and flossing around the surgical site until removal of the periodontal dressing. Information about eating and oral hygiene after the operation was given to the patients both verbally and in written form. All the patients were prescribed a mouthwash containing 0.12% chlorhexidine gluconate (Klorhex®, Drogsan İlaçları, Türkiye) and analgesia containing paracetamol (Parol 500 mg, Atabey, İstanbul, Türkiye). It was recommended to use the mouthwash for 30 s, 3 times a day, for a week, and to take the analgesia according to pain status. The sutures were removed on the 14th day postoperatively and mechanical plaque control precautions were restarted. The patients were instructed to use the modified Stillman technique with a soft bristle toothbrush to brush their teeth.

Clinical evaluations and measurements

The primary outcome of the study was graft shrinkage. The secondary outcomes included gingival recession depth (RD), keratinized tissue height (KTH), probing depth (PD), clinical attachment level (CAL), postoperative oedema and erythema, and patient-reported outcomes including pain scores and analgesic consumption.

Determination of the graft harvesting area

Photographs taken during the operation and at postoperative 1 month (T1), 3 months (T3), and 6 months (T6) time-points in the postoperative period were used to calculate the total graft area (Fig. 3). A periodontal probe (Hu-Friedy, Chicago, IL, USA) was used as reference for image calibration. When taking the clinical photographs, the patients were seated upright and looking straight ahead, and all the photographs were taken by the same person using the same camera (Nikon Coolpix 4500, Tokyo, Japan) at the same distance (20 cm) with the same light settings (ISO-800). All the measurements were performed on a high-resolution monitor at x10 magnification with specific software (National Institutes of Health, USA, Image J 1.48 V) (Fig. 4). When determining the graft area (mm2), borders were drawn and the area was calculated; each measurement was repeated three times and the average value was used in the analyses [7]. Calibration of the examiner was made through evaluation of measurements repeated at a 24-hour interval of the photographs of 10 patients not included in the study, and the ICC for the graft area was found to be 0.978. Shrinkage in the graft area was calculated taking into consideration the difference between graft area measurements at 1, 3, and 6 months from the baseline measurements. The shrinkage percentage at each follow-up time-point was calculated using the following formula: Shrinkage % = [(baseline surface area - follow-up time-point surface area)/baseline surface area x 100].

Clinical measurements

The clinical measurements were performed at 1 week preoperatively (T0) and at T1, T3, and T6. The gingival RD was defined as the distance from the CEJ to the gingival edge, KTH as the distance from the gingival edge to the mucogingival line, PD as the distance from the gingival edge to the sulcus layer, and CAL as the distance from the CEJ to the sulcus base. All the measurements were taken from mid-fascial points using a manual



Fig. 4 Clinical healing of the free gingival graft in the Control, LLLT, and i-PRF groups

periodontal probe and the values were rounded to the nearest millimetre [18]. RD and CAL were measured at all the time points, and PD and KTH only at the first and sixth months. Intra-rater reliability was evaluated by repeating the RD and PD measurements after one week in 10 patients who were not included in the study and the Intraclass Correlation Coefficient (ICC) was found to be 0.945 for RD and 0.925 for CAL. Changes occurring in the 6th month compared to baseline were calculated for PD, CAL, RD, and KTH, and at the end of the 6th month the root coverage percentage was determined.

Oedema and erythema in the recipient region

On postoperative days 3, 5, 7, 14, 21, and 28, oedema and erythema in the recipient area were evaluated and scored. Oedema was scored as 0 = absent or none, no inflammation; 1 = mild, intraoral oedema limited to the surgical site; 2 = moderate, extraoral oedema limited to the surgical site; 3 = significant, extraoral oedema spreading beyond the surgical site; 4 = severe, extraoral oedema spreading beyond the surgical site and limited mouth opening. The scoring of erythema in the surgical area was made as follows: 0 = absent or none, all tissues are pink; 1 = mild, < 25% of tissues are red; 2 = moderate, 25–50% of tissues are red; 3 = significant, 50–75% of tissues are red, 4 = severe, > 75% of tissues are red [7]. The calibration of the researcher taking the measurements was performed through repeated oedema and erythema scoring during the clinical examinations of 10 patients who had undergone FGG operation but were not included in the study. Intra-rater reliability was evaluated with the Kappa-Cohen test, and the values were calculated as κ = 0.918 for oedema, and κ = 0.896 for erythema.

Patient evaluation of postoperative morbidity

Each patient was instructed to record their daily pain levels and the number of analgesics taken during the first postoperative week using a provided form. Pain assessment was performed using a visual analog scale (VAS), with scores ranging from 0 (no pain) to 10 (maximum tolerable pain) [32]. The patients were also asked to record pain specifically at the recipient site.

Statistical analysis

The data obtained in the study were analyzed using IBM SPSS Statistics vn. 27 software (IBM Corp., Armonk, NY, USA). The distribution of continuous variables was assessed using the Shapiro–Wilk test and visual methods (histograms and Q–Q plots). Normally distributed data were presented as mean ± standard deviation values, and non-normally distributed data as median (minimum–maximum) values.

In this study, which included three groups and repeated measurements at multiple time points, repeated-measures ANOVA was used to evaluate the effects of time (within-subjects factor), group (between-subjects factor), and the group × time interaction. The sphericity assumption was assessed using Mauchly's test; when violated, the Greenhouse–Geisser correction was applied and results were reported using adjusted degrees of freedom.

The analyses primarily evaluated the main effects (time and group) and the group × time interaction. In

Table 1 Demographic characteristics

Age $_{\text{Mean}\pm\text{S.D}}$	Control Group	LLTL Group	i-PRF Group	*p*
	28.94 ± 4.88	**30.53 ± 5.46**	**27.59 ± 7.61**	[1]**0.408**
Sex $_{\text{n (\%)}}$				
Female	11 (%68.8)	12 (%80)	13 (%76.5)	[2]0.842
Male	5 (%31.2)	3 (%20)	4 (%23.5)	

*$p < 0.05$. [1] One-way ANOVA Test. [2] Fisher Freeman Halton Exact test

LLTL Low-Level Laser Therapy, *i-PRF* injectable Platelet-Rich Fibrin, *SD* Standard deviation

cases where a significant interaction or main effect was detected, intergroup comparisons across time points and within-group comparisons over time were performed. Bonferroni correction was applied to control the Type I error rate in multiple comparisons.

For intergroup comparisons at a single time point, one-way ANOVA was used when the normality assumption was met, and the Kruskal–Wallis test when it was not. For within-group comparisons between two time points, the Paired Samples t-test was used for normally distributed data, and the Wilcoxon signed-rank test for data not showing normal distribution.

In addition to graft area measurements, percentage shrinkage values reflecting the normalized change relative to baseline were also calculated. This parameter was evaluated separately in the analyses, as it provides a more sensitive measure for intergroup comparisons by accounting for individual baseline differences.

All statistical analyses were performed with the level of statistical significance set at $p < 0.05$.

Results

The study included a total of 48 patients, comprising 36 (75%) females and 12 (25%) males in the age range of 19–41 years. The demographic characteristics of the groups are shown in Table 1.

No significant difference was seen between the groups in respect of PD at T0 and T6 ($p > 0.5$). In the control and i-PRF groups, no significant change was seen in the PD level from baseline to T6 ($p > 0.05$). In the LLLT group, the PD level was observed to be significantly reduced at T6 compared to T0 ($p = 0.008$). There was no significant difference between the groups in respect of CAL at T0 ($p > 0.05$). The T6 CAL values showed a sigificant difference ($p = 0.010$). At T6, the CAL was significantly lower in the i-PRF group than in the control group ($p = 0.007$). At T6, the CAL values were determined to be significantly decreased compared to T0 in all three groups ($p = 0.001$) (Table 2).

The sphericity assumption for repeated measures ANOVA regarding KDH, RD, and graft area was evaluated using Mauchly's test, and a violation was detected ($p < 0.001$); therefore, the Greenhouse–Geisser correction was applied. After correction, the time effect was found to be statistically significant for all three parameters ($p < 0.001$). A significant group × time interaction was observed for KDH and RD ($p = 0.004$ and $p < 0.001$, respectively), and an overall significant difference between groups was also detected ($p < 0.001$). In comparisons across time points, significant differences between groups were found at T1, T3, and T6 ($p < 0.05$); the i-PRF group exhibited higher KDH values compared to the control group, while RD values were significantly lower than both the control and LLDL groups ($p < 0.05$). The group × time interaction for graft area was determined to be statistically significant ($p < 0.001$), with no significant differences between groups at individual time points or in overall mean values ($p > 0.05$; $p = 0.350$). A significant reduction in graft area over time was observed in all groups ($p < 0.05$) (Table 3).

For percentage graft shrinkage, the sphericity assumption was violated according to Mauchly's test ($p < 0.001$), and the Greenhouse–Geisser correction was applied. The time effect and intergroup differences were statistically significant ($p < 0.001$), whereas the group × time interaction was not significant ($p = 0.227$). Significant differences between groups were observed at T1, T3, and T6 ($p < 0.05$); the lowest percentage shrinkage values were found in the i-PRF group, while the LLDL group also showed lower percentage shrinkage compared to the control group ($p < 0.05$). Within-group analyses revealed a significant decrease in percentage shrinkage over time

Table 2 Intra-group and inter-group evaluation of Buccal PD and CAL

		Control Group		LLDL Group		i-PRF Group		
		Mean ± S.D.	**Median (Min-Max)**	**Mean ± S.D**	**Median (Min-Max)**	**Mean ± S.D**	**Median (Min-Max)**	***p***
PD	T0	2.25 ± 0.68	2 (1–3)	2.33 ± 0.62	2 (1–3)	2.18 ± 0.53	2 (1–3)	[1]0.719
	T6	1.94 ± 0.44	2 (1–3)	1.87 ± 0.35	2 (1–2)	1.94 ± 0.43	2 (1–3)	[1]0.860
	[2]p	0.059		0.008*		0.157		
CAL	T0	7.44 ± 0.81		7.5 ± 0.74		7.42 ± 0.77		[3]0.955
	T6	3.43 ± 0.53		3.17 ± 0.48		2.86 ± 0.52		[3]0.010*
	[4]p	0.001*		0.001*		0.001*		

*$p < 0.05$. [1]Kruskal Wallis Test. [2]Wilcoxon Sign test. [3]Oneway ANOVA Test.[4]Paired samples t test

T0 1 week preoperatively, *T6* 6th month postoperatively, *LLTL* Low-Level Laser Therapy, *i-PRF* injectable Platelet-Rich Fibrin, *PD* Probing depth, *CAL* Clinical attachment level, *SD* Standard deviation. Since the CAL parameter showed a normal distribution, the data are presented as mean ± standard deviation, and median and minimum–maximum values were not reported

Table 3 Intra-group and inter-group evaluation of KTH, RD, and graft area

		Control Group	LLDL Group	i-PRF Group	
		Mean ± S.D.	Mean ± S.D.	Mean ± S.D.	[1]p
KTH	T0	1.29 ± 0.39	1.19 ± 0.40	1.22 ± 0.45	0.788
	T1	4.13 ± 0.90	4.35 ± 0.59	4.94 ± 0.69	0.009*
	T3	4.01 ± 0.84	4.29 ± 0.54	4.88 ± 0.71	0.003*
	T6	3.94 ± 0.85	4.25 ± 0.52	4.85 ± 0.73	0.003*
	[2]p	0.001*	0.001*	0.001*	
RD	T0	5.19 ± 0.38	5.17 ± 0.57	5.24 ± 0.44	0.899
	T1	1.61 ± 0.45	1.39 ± 0.43	0.97 ± 0.29	0.001*
	T3	1.55 ± 0.41	1.34 ± 0.40	0.94 ± 0.31	0.001*
	T6	1.49 ± 0.37	1.30 ± 0.37	0.92 ± 0.31	0.001*
	[2]p	0.001*	0.001*	0.001*	
Graft area	Baseline	87.51 ± 18.58	89.77 ± 15.5	88.52 ± 15.83	0.931
	T1	68.79 ± 14.39	74.71 ± 11.88	77.27 ± 14.11	0.197
	T3	69.58 ± 14.98	76.37 ± 13.07	78.57 ± 14.15	0.178
	T6	70.10 ± 14.41	78.01 ± 13.24	79.74 ± 14.09	0.123
	[2]p	0.001*	0.001*	0.001*	

*$p < 0.05$. [1]Oneway ANOVA Test. [2]Repeated Measures Analysis of Variance

T0 1 week preoperatively, *T1* 1st month postoperatively, *T3* 3rd month postoperatively, *T6* 6th month postoperatively, *LLTL* Low-Level Laser Therapy, *i-PRF* injectable Platelet-Rich Fibrin, *PD* Probing depth, *CAL* Clinical attachment level, *KTH* Keratinized tissue height, *RD* Recession depth, *SD* Standard deviation

Table 4 Intra-group and inter-group evaluation of shrinkage amount and root coverage percentage

		Control Group	LLDL Group	i-PRF Group	
		Mean ± S.D.	Mean ± S.D.	Mean ± S.D.	[1]*p*
Shrinkage amount (%)	T1	21.32 ± 2.21	16.56 ± 3.38	12.75 ± 1.67	0.001*
	T3	20.53 ± 1.56	14.85 ± 3.66	11.22 ± 1.3	0.001*
	T6	19.79 ± 2.12	13.03 ± 2.61	9.85 ± 1.6	0.001*
	[2]p	0.058	0.001*	0.001*	
Root coverage percentage (%)		71.0 ± 7.74	74.31 ± 8.79	82.15 ± 6.46	0.001*

*$p < 0.05$. [1]Oneway ANOVA Test. [2]Repeated Measures Analysis of Variance

T1 1st month postoperatively, *T3* 3rd month postoperatively, *T6* 6th month postoperatively, *LLTL* Low-Level Laser Therapy, *i-PRF* injectable Platelet-Rich Fibrin, *SD* Standard deviation

in the i-PRF and LLDL groups ($p < 0.05$), whereas no significant change was observed in the control group ($p > 0.05$). Percentage shrinkage values, which normalize baseline differences, provided greater discriminative power in intergroup comparisons (Table 4).

The changes in T0-T6 PPD, RD, CAL, and graft area are shown in Table 5. There was no difference between the groups in the changes in PD and CAL ($p > 0.05$). The increase in KTH was found to be greater in the i-PRF group than in the control group ($p = 0.006$). The decrease in recession depth was greater in the i-PRF group than in the control group ($p = 0.023$). The decrease in graft area was determined to be greater in the control group than in the LLLT group ($p = 0.009$) and the i-PRF group ($p = 0.001$).

Postoperative oedema, erythema, pain, and use of analgesia are shown in Fig. 5. No significant difference was determined between the groups in respect of oedema and erythema on days 1, 3, 5, 7, 14, 21, and 28 and the VAS scores on days 1–7 ($p > 0.05$). Analgesia use on postoperative days 2, 3, and 5 was seen to be significantly higher in the control group than in the i-PRF group ($p < 0.05$). In the other group comparisons and on other days, no statistically significant difference was determined ($p > 0.05$).

Discussion

In this study, the effects on graft dimensions, root surface coverage rate, and postoperative patient comfort of the application of i-PRF over FGG was compared for the first time with LLLT and control groups. The study findings demonstrated that i-PRF application significantly improved postoperative clinical outcomes by decreasing graft shrinkage and increasing root surface coverage. The decrease in RD and increase in KTH were more significant in the i-PRF group than in the control and LLLT groups. Graft shrinkage was observed to be less in the LLLT group than in the control group, but compared to the i-PRF group, these effects remained limited. Although a decrease was observed in the oedema, erythema, and postoperative pain scores in the i-PRF and LLLT groups, these differences were not found to be

Table 5 Evaluation of the groups in terms of changes from T0 to T6

	Control Group		LLDL Group		i-PRF Group		
	Mean ± S.D.	Median (Min/Max)	Mean ± S.D.	Median (Min/Max)	Mean ± S.D.	Median (Min/Max)	*p*
PD	-0.31 ± 0.6	0 (-1/1)	-0.47 ± 0.52	0 (-1/0)	-0.24 ± 0.66	0 (-1/1)	0.612
CAL	-4.01 ± 0.88	-4 (-6/-2.7)	-4.33 ± 0.9	-4.4 (-5.8/-2.8)	-4.55 ± 1.01	-4.5 (-6.2/-2.3)	0.230
KTH	2.65 ± 0.86	2.7 (1.3/4)	3.05 ± 0.55	3.1 (2.1/3.9)	3.62 ± 0.83	3.8 (2/5)	0.009*
RD	-3.7 ± 0.58	-3.8 (-5/-2.8)	-3.87 ± 0.73	-4 (-4.9/-2.1)	-4.32 ± 0.6	-4.3 (-5.6/-3.3)	0.026*
Graft area	-17.41 ± 4.77	-16.1 (-29/-10.3)	-11.77 ± 3.44	-11 (-19.1/-7)	-8.79 ± 2.23	-9.3 (-12/-4.3)	0.001*

Kruskal Wallis Test, *$p < 0.05$

T0 1 week preoperatively, *T6* 6th month postoperatively, *LLTL* Low-Level Laser Therapy, *i-PRF* injectable Platelet-Rich Fibrin, *PD* Probing depth, *CAL* Clinical attachment level, *KTH* Keratinized tissue height, *RD* Recession depth, *SD* Standard deviation

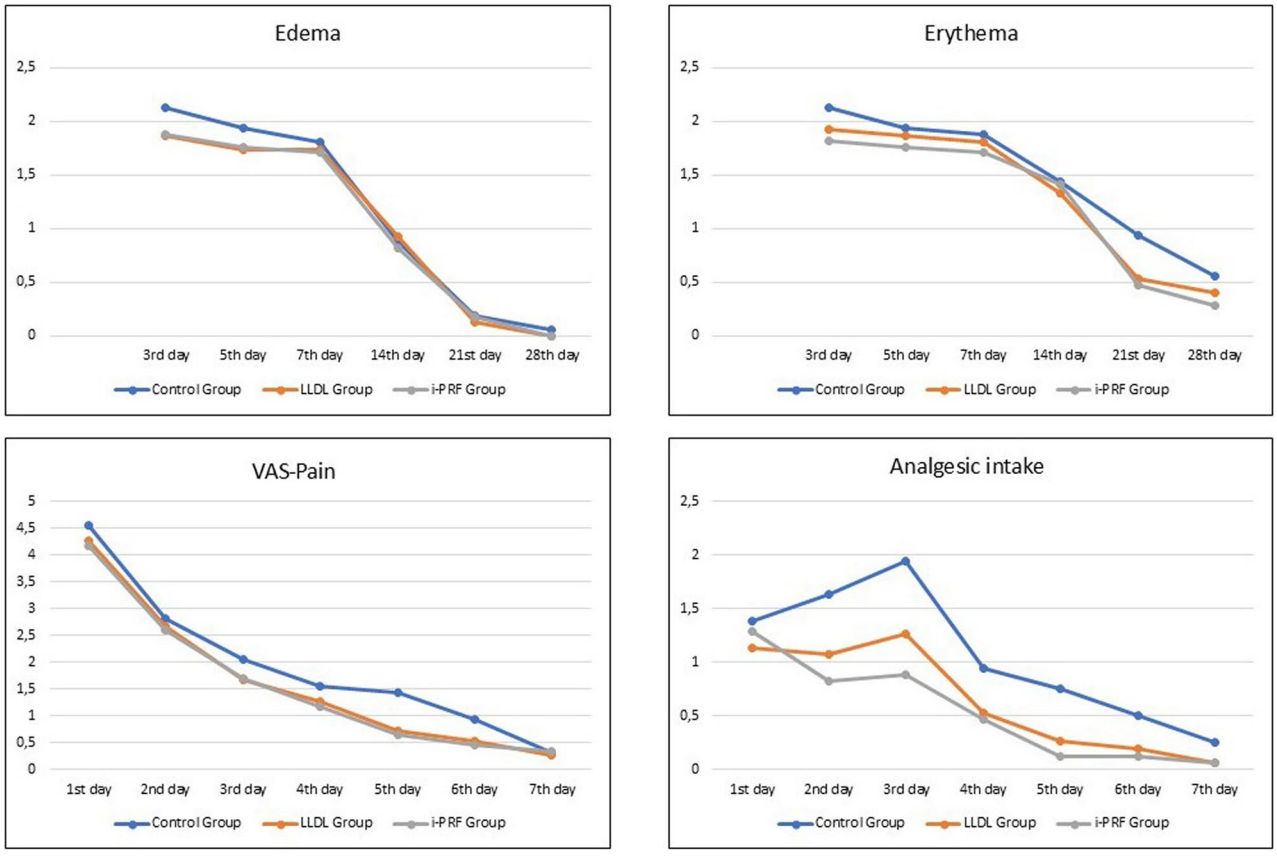


Fig. 5 Postoperative oedema, erythema, pain, and analgesic intake

statistically significant. Postoperative analgesic use was lower in the i-PRF group than in the control group. These results showed that the null hypothesis was rejected in respect of i-PRF and partially accepted in respect of LLLT.

In this study, the application of i-PRF gel over FGG improved clinical outcomes by reducing graft dimensional shrinkage compared to the LLLT and control groups. Moreover, in the groups where i-PRF gel was applied over FGG, KTH and root coverage were significantly higher than in the control and LLLT groups. These findings suggest that i-PRF may have positive effects on graft stability and clinical success in FGG procedures. In a study by Amr, the application of a PRF membrane beneath FGG was reorted to reduce horizontal and vertical dimensional changes of the graft in the early healing period compared to FGG alone [33]. Furthermore, another study demonstrated that the injection of i-PRF onto the root surface as a biomodification agent during FGG surgery significantly increased the rate of root coverage [23]. Uçak et al. reported that applying i-PRF gel as a three-dimensional scaffold or "pouch" for CTG may enhance vascularization and facilitate the transfer of growth factors [18]. These studies are consistent with the current study findings and support the notion that i-PRF may positively influence graft stability and healing processes.

In a study evaluating donor site healing [34], PRF application was reported to increase mucosal thickness in the first week, provide mechanical support through its fibrin network, and accelerate neoangiogenesis [35]. These findings appear to be consistent with the improved clinical healing observed in the i-PRF group in the current study. i-PRF has been reported to be rich in growth factors such as PDGF-AA, PDGF-AB, EGF, and insulin-like growth factor-1 (IGF-1), and these factors can be released over a longer period via the fibrin matrix [21, 36]. This sustained release may contribute to cell proliferation, neoangiogenesis, and connective tissue formation, thereby influencing the clinical outcomes observed. However, these mechanisms were not directly evaluated in the current study, and the findings should therefore be interpreted in the light of the existing literature.

In this study, the application of LLLT at a wavelength of 940 nm and an energy density of 5 J/cm² to FGG significantly reduced graft shrinkage compared to the control group. Although a positive trend in favour of LLLT was observed in terms of root coverage and KTH, these differences did not reach statistical significance. These findings suggest that under the specific LLLT protocol used,

there may be some beneficial effects on graft stability. The results are partially consistent with findings reported in the literature. In a study by Mehmet and Günpınar, LLLT applied at 810 nm with an energy density of 6 J/cm² was reported to reduce graft shrinkage following FGG. However, it should be considered that the effectiveness of LLLT may vary depending on the wavelength, energy density, and application protocol, and different protocols may yield different outcomes. Several studies have reported that LLLT can influence cellular proliferation and healing processes at the cellular level [29, 37, 38].

There is still limited and conflicting evidence in the literature regarding the clinical efficacy and optimal parameters of LLLT. The energy density should be sufficient to stimulate biological activity without causing tissue damage; according to the Arndt–Schulz law, low doses may be ineffective, whereas high doses may exert inhibitory effects [38]. In this context, an energy density of 5 J/cm² was selected in the current study. Tissue penetration of laser light depends on the wavelength, and LLLT is generally applied within the range of 600–1100 nm. It has been reported that red light penetrates more superficial tissues, whereas near-infrared light reaches deeper tissues [29]. These characteristics indicate that the applied parameters may play a decisive role in the biological response. In conclusion, the results of the current study suggest that LLLT may support graft healing following FGG, although its effectiveness appears to be dependent on the applied protocol. The lack of a standardized protocol and variability in parameters should be taken into account when interpreting these results.

Changes can develop in FGG dimensions during the healing period. Different clinical studies have reported rates of shrinkage in the graft surface area (mm^2) varying between 12% and 58% [7, 8, 39–41]. In a recent study, Monje et al. performed graft measurements with the periodontal probe method and observed 33% shrinkage [10]. Using digital callipers for the measurements, Silva et al. reported high shrinkage rates with a decrease in FGG surface area of 58% in smokers and 44% in non-smokers [40]. In the current study, standardised clinical photographs and a Java-based photogrammetric analysis method were used to measure the graft surface area and the shrinkage rates were found to be between 9% and 21%; these values were consistent with the literature. Although it has been reported in the literature that graft shrinkage can continue for up to 6 months after FGG [9, 39], a significant increase was observed in graft surface area at 3 and 6 months in the current study LLLT and i-PRF groups. In contrast to other studies in the literature, the graft surface area remained stable after 1 month in the control group. These results demonstrate that computer-based photogrammetric analysis provided a more sensitive and reliable evaluation than the traditional probe or calliper measurements. However, inter-personal differences may be due to recipient area tissue phenotypes and this could affect surface area changes with graft adaptation.

The healing process of FGG is shaped by the biological integration of the graft with the recipient site. Previous studies have shown that graft viability initially depends on avascular plasmatic diffusion from the surrounding tissues, followed by capillary formation and the development of anastomoses between the recipient bed vessels and the graft tissue [42]. This vascularization process is critical for graft survival and stabilization; therefore, supporting blood flow and the presence of growth factors during the early healing phase are among the key factors that may influence graft success. However, it has been suggested that interposition placement of PRF in membrane form may limit the revascularization process by creating clot thickness [43]. Although Amr et al. reported that placing PRF beneath FGG contributed to the preservation of graft dimensions [33], considering this potential limitation, i-PRF was applied directly over the FGG in the present study, without creating a barrier between the graft and the recipient bed.

Although the current study results showed that the oedema, erythema, and pain scores were lower in the LLLT and i-PRF groups than in the control group in the early healing period, these differences were not found to be statistically significant. Yildiz et al. reported that LLLT after FGG had no significant effect on oedema [7]. Scott et al. showed that PRF applications made no significant difference in respect of redness and oedema [44]. These results are consistent with the current study findings. The control and LLLT groups in the current study reported a significantly higher number of analgesics used in the first week than the i-PRF group. Although the lower need for analgesia of the i-PRF group was consistent with previous studies [26], the difference has been reported not to be statistically significant in some studies [18, 44]. Differences in pain and analgesia consumption may be due to personal differences in biological response, limited sample size, the type of thrombocyte concentrates used, pain management protocols, and variables in the subjective evaluation methods. Similarly, although there are studies showing that LLLT decreased pain and the need for analgesia, there are also studies reporting that laser application has no significant effect on postoperative pain scores. As the effects of LLLT and i-PRF are primarily at the cellular level, it can be expected that clinical differences are not evident in the early postoperative period. The clinical efficacy of LLLT depends on the interaction of the laser beam with the target tissue, and parameters such as wavelength, energy intensity, duration of irradiation, treatment intervals, and application distance can significantly affect the outcome. There is currently

no standard, well-defined protocol for LLLT and the parameter selection is usually based on the clinician's experience and preference. It should also be taken into consideration that the analgesics used in the postoperative period could have an effect on the results.

In the current study, although no statistically significant difference was observed between the groups in terms of the absolute graft area during the follow-up period, significant differences were detected in percentage shrinkage. This finding can be explained by the normalization of percentage change relative to baseline values. Percentage shrinkage provides a relative assessment by accounting for individual variations in baseline graft dimensions and may more accurately reflect graft stability. In contrast, absolute area measurements represent only a single time-point assessment. Therefore, percentage shrinkage emerges as a more appropriate parameter for evaluating temporal changes in the graft.

In the LLLT group, a statistically significant reduction in PD values was observed. However, the mean change of 0.46 mm suggests that the clinical relevance of this difference may be limited. This finding indicates that LLLT may have a statistically significant but clinically minimal effect on PD under the protocol used in this study. Nevertheless, it should be considered that the therapeutic efficacy of LLLT may vary depending on the applied parameters, and these results should be interpreted in relation to the specific protocol used.

The i-PRF used in the current study was applied homogeneously to the FGG surface without delay following the completion of the expected polymerization period. Taking advantage of the adhesive properties of i-PRF [22], a stable layer was formed on the FGG surface, and this stability was further supported using sterile aluminum foil and a periodontal dressing. Throughout the study period, no early dislodgement of the periodontal dressing was observed in any patient.

This study had several limitations. Due to the parallel-group design, individual variations in healing may have influenced the results. Although an attempt was made to standardize graft thickness to approximately 1.5–2 mm using a periodontal probe, neither graft thickness nor the phenotype of the recipient site was objectively measured using digital callipers. Clinical measurements were performed manually using a periodontal probe without any standardized measuring device. However, all measurements were taken by a single calibrated and blinded examiner, which improved standardization but may still be considered a methodological limitation. Photographs were obtained with positioning and angles standardized as far as possible. However, the absence of a tripod or mechanical positioning device may have introduced minimal variations between measurements. In addition, the use of different suture techniques in a limited number of cases, although evenly distributed among groups, may be considered a minor methodological variation. Pain is a subjective sensation that is difficult to define, measure, and evaluate, and patients may have difficulty distinguishing between pain in the donor and recipient sites. To minimize this potential bias, patients were repeatedly instructed to differentiate between pain in the donor and recipient sites, and only pain originating from the recipient site was included in the evaluation. Nevertheless, it cannot be ruled out that patients may have inadvertently included pain related to the donor site in their assessments. These limitations necessitate cautious interpretation of the findings and highlight the need for further studies with longer follow-up periods, improved standardization, and inclusion of biological parameters.

Conclusion

In conclusion, the results of this study showed that i-PRF application in FGG operations increased root surface coverage by reducing graft shrinkage and improved postoperative patient comfort. These findings suggest that the use of i-PRF as a biological support material in FGG surgeries could make clinically significant contributions to the healing process. In contrast, although LLLT contributed to graft stability, it demonstrated a more limited effect compared to i-PRF under the specific protocol used in this study. Since the efficacy of LLLT may vary depending on the applied protocol, further standardized studies are required.

Supplementary Information

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Supplementary Material 1.

Supplementary Material 2.

Supplementary Material 3.

Supplementary Material 4.

Supplementary Material 5.

Authors' contributions

Methodology: [Esra BOZKURT], [Eda ÇETIN ÖZDEMIR]; conceptualization: [Esra BOZKURT]; Material preparation: [Esra BOZKURT]; data collection: [Esra BOZKURT]; analysis: [Esra BOZKURT], [Eda Çetin Özdemir]; Literature search: [Esra BOZKURT]; supervision: [Esra BOZKURT], [Eda ÇETIN ÖZDEMIR]. The first draft of the manuscript was written by [Esra BOZKURT], and all authors commented on previous versions of the manuscript. All authors read and approved of the final manuscript.

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Data availability

Data and materials are available at the Periodontology Department in the Faculty of Dentistry, Kahramanmaraş Sütçü Imam University.

Declarations

Ethics approval and consent to participate

All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional review board of Clinical Research Ethics Committee of Inonu University (protocol no: 2024/84) and with the 2013 Helsinki declaration and its later amendments or comparable ethical standards. This study was retrospectively registered at ClinicalTrials.gov (Identifier: NCT07303192; Date of registration: 2025-12-11). Informed consent was obtained from all participants individually included in the study.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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